MUBBY DIMEJI

Senior Data Engineer

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Nationality .... *US Citizen*

LINKS

GtiHub, LinkedIn

PROFILE

Hi, I'm Mubby. I've been a data engineer for about 10 years now, mostly in fintech. I got my start at Accenture, where every week felt like a new puzzle, one day I was building ML models to predict when machines would break, the next I was putting together a chatbot for an airline. It was like boot camp for curiosity, and it shaped how I look at problems even now. These days I'm at Varobank, building the big data systems that keep money flowing around the world. What keeps me hooked is the challenge, fraudsters getting smarter, data getting messier, systems needing to move faster. I see those problems like a wrestling match. I'll circle them, untangle them, push against them until they finally give way. For me, that's the fun of it. And when the result is a small business finally being able to accept payments, or someone sending money home without stress, it makes every late night worth it. I'm always chasing the next thing to learn. Right now, it's NLP, the idea of teaching machines to not just crunch numbers but actually understand language. To me, that's the kind of puzzle that makes this field exciting. At the end of the day, I'm just someone who thrives on solving problems, whether it's an ugly dataset, an over-complicated query, or building something people can actually feel in their everyday lives.

WORK EXPERIENCE

Senior Data Engineer, Varo Bank .... Sep 2022 — Present

Real-time fraud signals from card transactions – built from scratch to reduce detection lag and increase signal speed

- Fraud and risk teams at Varo needed transaction data to arrive in real time to stop fraud before a purchase was completed. If feeds lagged, fraud detection became forensic cleanup instead of prevention.
- Built the real-time ingestion system around Marqeta, Varo's card processor responsible for card issuance, authorization, and settlement. Integrated with Marqeta's Transactions API and Just-in-Time Funding APIs, which allow Varo to approve or decline payments in flight.
- Developed ingestion in Python with authentication, retries, and session management. Added a resilience layer with exponential backoff for unstable environments, circuit breakers to stop cascading failures, and monitoring to detect throughput or queue slowdowns.
- Implemented stepped retry delays starting at one second and doubling up to sixteen seconds to respect Marqeta's rate limits and avoid sandbox lockouts.
- Routed ingested events through Debezium change data capture into Kafka, applied schema compatibility checks, then loaded into Snowflake in small batches with a custom low-latency loader that could recover gracefully from failed inserts.
- Automated secret management with AWS Secrets Manager, rotated quarterly using Lambda, and dynamically updated Airflow DAGs through Helm. Documented the lifecycle in an internal Secret Rotation Tracker to make adoption easy for other teams.
- Created a synthetic data process to inject fake card swipes for regression testing. Fraud teams used it to validate new rules safely. Extended the pipeline to include new fraud features such as transaction count statistics and velocity indicators to improve models.
- Delivered this proactively after a risk posture review flagged the lack of real-time data as a gap. Outcome: near-real-time fraud visibility that allowed the business to block fraud while it happened rather than reimburse it afterwards.

Unified platform for daily partner data feeds – built to guarantee next-day marketing and revenue insights

- Marketing and growth teams relied on partner feeds (e.g., advertising, leads, income verification), but these were inconsistent, late, or malformed, regularly breaking reporting.
- Built a unified ingestion framework in Airflow and Glue to standardize landing, validation, and promotion of partner feeds. Expanded it to run a dozen feeds including lead files and income verification data without requiring custom pipelines for each vendor.
- Landed files into S3 staging areas and validated them with Great Expectations, enforcing rules such as: must contain at least one row, fewer than ten percent nulls, and correct data types.
- Promoted validated files into live storage using Supernova, Varo's internal wrapper around AWS copy operations, which resets metadata and access controls consistently.
- Built monitoring to detect when data was missing for more than three days and tied this to Airflow retries, eliminating silent feed failures.
- Put all DAGs under version control with schema-difference checks and continuous integration pipelines that blocked unsafe changes.
- Delivered a schema change that added a new marketing metric so the growth team could measure campaign lift directly without manual work.
- Automated schema provisioning with a templated script that created new schemas with analyst read-only access and QA mirrors kept in sync by Jenkins. This removed the need for access tickets.
- Outcome: Partner data became dependable, and Finance and Marketing could close spend reports by 10 a.m. every day, meeting next-day SLA without manual reconciliation.

Airflow logging and alerting redesign – removed noise, clarified signals, and improved operational response

- Airflow was the backbone of data workflows but its logging and alerting were brittle. Logs were sometimes lost during worker restarts, alerts were too generic, and engineers wasted hours chasing phantom failures.
- Redesigned logging to stream directly into S3 and Elastic for durability and searchability. Added a fallback log handler and documented recovery steps in a runbook.
- Rebuilt alerting around the internal Severity and Operational Alerting (SAOA) framework: critical jobs triggered PagerDuty, high-severity jobs triggered Slack alerts, and informational alerts were suppressed.
- Enriched alerts with run IDs, DAG links, and direct log URLs to allow immediate triage.
- Fixed environment drift by upgrading staging from Airflow 2.1 to 2.4 to match production. The upgrade required migrating DAGs, updating operators, rebuilding containers, and re-testing dependencies. This aligned environments, enabled newer workflow syntax, and eliminated reproducibility problems.
- Outcome: On-call engineers now receive targeted, context-rich alerts. Logs are always available, and environment parity means issues can be reproduced and fixed consistently.

Shipped data protections for PCI and GDPR – cleared compliance blockers and launched safely into production

- Compliance required proof of PCI-DSS and GDPR adherence before cardholder analytics pipelines could go live.
- Implemented a masking transform to irreversibly hash card numbers before they reached Snowflake, applying to both new data and historical records.
- Added a GDPR retention policy: Airflow workflows deleted transaction records older than 400 days, with logs and audits reviewed by Security before launch.
- Outcome: Pipelines cleared compliance checks, production launch was unblocked, and the business gained provable PCI/GDPR evidence.

Improved secrets rotation, access governance, and alert observability across the platform

- Automated rotation for all sensitive credentials (partner feeds, Marqeta, internal APIs) using AWS Secrets Manager, Lambda, and Helm injection into Airflow. Published the lifecycle in the internal Secret Rotation Tracker.
- Partnered with IAM to align Okta single sign-on groups with AWS IAM roles, cutting analyst onboarding time from days to hours and eliminating dangling permissions.
- Extended Airflow alert enrichment so all pipeline notifications included the cause and direct logs, rather than just saying "task failed."
- Outcome: Secrets rotated routinely, onboarding accelerated, and alert triage became quick and reliable.

Delivered time-partitioned campaign data for linear TV – made marketing performance traceable to hour blocks

- A media partner sent daily flat files of TV advertising spend, but marketing needed hour-by-hour breakdowns to know which slots were effective.
- Added a time-slicing step in Airflow to split daily files into hourly partitions and load into Snowflake.

- Cleaned up legacy schemas so partitions aligned with reporting tools, reducing analyst confusion.
- Outcome: Marketing gained ROI visibility per broadcast slot and reallocated spend away from low-performing hours toward high-performing ones.

Prototyped Autoloader ingestion to inform the next wave of platform modernization

- Glue jobs were costly and batch-oriented, so I tested Databricks Autoloader as a streaming alternative.
- Built a prototype pipeline for an income verification feed with identical scheduling to Glue, injected schema drift scenarios, validated completeness, and compared cost and performance.
- Results showed Autoloader handled schema changes better and reduced cost. Findings were presented to Platform Engineering to shape the roadmap.
- Outcome: Provided a scoped, data-driven evaluation that informed budget and migration decisions.

Cross-team collaboration and long-term ownership

- Led weekly ingestion syncs with Fraud Strategy, Marketing Analytics, Security, and SRE to align launches, validate feeds, and resolve ownership issues.
- Acted as the escalation point during pipeline incidents and primary engineer for partner onboarding and DAG deployments.
- Mentored two junior engineers on Airflow and validation, then transitioned ownership of DAGs once they were confident.
- Documented everything, ensured reproducibility, and eliminated reliance on tribal knowledge or one-off fixes.

Machine Learning Engineer, Accenture .... Jan 2017 — Sep 2021

Predictive Maintenance for Manufacturing – Built a large-scale system to predict machine failures

- Designed a predictive maintenance system for a major auto parts manufacturer, ingesting sensor data from over 10,000 machines including assembly robots, CNC machines, and conveyors.
- Set up a 50-node Hadoop cluster on-premises, configured YARN with separate resource pools for ETL and analyst ad-hoc queries, and tuned it so heavy crunching jobs did not slow down analysts running quick checks.
- Partitioned Hive tables by timestamp and machine type to accelerate queries on readings and maintenance logs.
- Developed PySpark pipelines to clean noisy sensor data, interpolate missing values, and correct sensor drift using median filters.
- Engineered features such as vibration level averages, Fourier transforms on temperature signals, lag features between sensor readings, and frequency-domain features from vibration data to spot failing bearings.
- Built anomaly detection models with Isolation Forests to capture unusual power spikes and abnormal vibrations, and Random Forests to predict known issues such as bearing wear or gear misalignment.
- Combined models into a hierarchical approach: base models covered general metrics like power consumption, while expert models targeted specific machine types such as robotic arms.
- Trained models with MLlib, tracked experiments and comparisons in MLflow, and tuned Random Forest and Isolation Forest results against test sets.
- Delivered daily 30-day risk scores and predicted failure modes for each machine, integrating results directly into the client's maintenance scheduling system.

Anomaly Detection in Financial Transactions – Detected fraud in real time at scale

- Built an anomaly detection system for a bank that processed millions of transactions daily, with each transaction described by hundreds of features such as amount, location, merchant type, and user behavior.
- Combined unsupervised methods (Isolation Forests distributed on Spark clusters) with supervised learning (XGBoost trained on labeled fraud data).
- Designed engineered features including graph-based relationships between users and merchants, aggregations over short time windows (e.g., number of transactions in the last hour), and longer-term trends (e.g., daily totals).
- Streamed transactions through a Kafka pipeline so each one was scored immediately by the models.
- Flagged high-risk transactions for human review while categorizing low-risk ones automatically.
- Deployed the system on-premises with Kubernetes for microservice orchestration, and monitored with Prometheus and Grafana.
- Outcome: Caught fraudsters who previously flew under the radar by comparing individual spending behavior against peer groups and historical patterns.

Time Series Forecasting for Retail Inventory – Optimized stock levels across 500 stores

- Created a demand forecasting system for a retail chain managing over 500 stores and tens of thousands of products.
- Built data pipelines using Azure Data Lake Storage, Databricks, and Spark for distributed computing.
- Applied the Prophet library to capture multiple seasonalities and holidays; customized Prophet to model unusual seasonal patterns (for example, swim gear spiking before summer and again during unexpected heat waves).
- Added external regressors including local events and weather.
- Developed a separate flash-sales model using features such as discount percentage, category, and historical promotion performance, integrated with social media sentiment analysis and Google Trends.
- Produced forecasts at multiple levels (store, category, SKU) and automated daily retraining and updates through Azure Data Factory.
- Achieved fast runtimes — full forecasts across all products and stores completed in about two hours on a 20-node Databricks cluster.
- Outcome: Improved forecast accuracy reduced both stockouts and overstocking, directly improving sales and lowering waste.

Computer Vision for Quality Control – Automated PCB defect detection

- Built a computer vision system for an electronics manufacturer to detect defects on printed circuit boards.
- Processed 4K resolution images of boards to identify defects such as solder bridges, missing components, and misalignments.
- Applied OpenCV for preprocessing (noise reduction, Gaussian blurring, thresholding under different lighting conditions, and morphological cleaning).
- Used SIFT feature extraction for component recognition and combined it with sliding-window SVM classifiers to locate and classify components.
- Built defect classifiers using SVMs tuned through experimentation with kernels and parameters; used data augmentation (rotations, lighting changes) to make the model robust.
- Trained models on annotated images and deployed them on GPUs in an on-premises data center, later migrating to NVIDIA Jetson devices at the factory floor for real-time inspection.
- Outcome: System became a critical part of QA, detecting defects reliably before boards shipped.

Natural Language Processing for Customer Service – Mined insights from millions of interactions

- Built an NLP system for a telecom client to analyze millions of call transcripts and chat logs.
- Designed a Spark pipeline for distributed processing: tokenization, stop-word removal, and TF-IDF vectorization.
- Applied Latent Dirichlet Allocation (LDA) for topic modeling, experimenting with priors and number of topics to balance interpretability with runtime.
- Combined lexicon-based sentiment analysis with supervised models: integrated VADER with a logistic regression model trained on over 100,000 labeled conversations, enriched with n-grams, sentiment lexicon features, and syntactic cues from CoreNLP.
- Outcome: Dashboards in Tableau displayed sentiment trends, recurring customer issues, and the effect of fixes. Helped the client uncover a major billing issue affecting thousands of customers.

Recommender System for Commerce – Personalized recommendations across millions of users

- Built a hybrid recommendation system that combined collaborative filtering with content-based filtering to personalize shopping experiences.
- Applied matrix factorization using ALS (Alternating Least Squares) to model user-item interactions at scale, while developing content-based features from product descriptions and categories using word embeddings and cosine similarity.
- Balanced ALS accuracy with content-based methods to handle cold-start problems for new users and products.
- Implemented offline computation of recommendations with Spark on Google Dataproc, storing results in BigQuery, and adjusted them in real time with Cloud Functions based on session behavior.
- Orchestrated the daily pipeline with Cloud Composer (managed Airflow).
- Outcome: Delivered a recommendation engine that scaled to millions of interactions, balancing accuracy with discovery of less obvious products to keep users engaged.

Chatbot Development for Airline Customer Service – Automated flight and booking assistance

- Built a customer-facing chatbot for an airline's service operations, handling queries such as flight status, booking changes, and baggage questions.
- Implemented a rule-based component with AIML for structured queries and developed a machine-learning NLU pipeline with spaCy for tokenization, part-of-speech tagging, and entity recognition.
- Fine-tuned a BERT model for intent classification and entity extraction to handle natural queries like "Change my flight from New York to London on July 15th."

- Designed a dialog management system combining a finite state machine with context-aware conversation tracking, enabling the bot to handle topic shifts and follow-up questions.
- Trained the model on thousands of real annotated conversations and integrated it with backend airline systems through RESTful APIs.
- Added frustration detection to escalate difficult interactions to human agents.
- Deployed on Azure Kubernetes Service with API Management for routing.
- Outcome: Reduced live-agent load while improving customer satisfaction through faster and more accurate self-service.

EDUCATION

Valparaiso University — Aug 2021

Master of Science in Analytics & Modelling — Indiana

Graduated with High Honors

SKILL SECTION

Data Engineering & Processing: Hadoop, Spark, Kafka, Airflow; ETL/data lakes with AWS Glue, Redshift, Azure Data Factory, Google BigQuery, Dataflow.

Cloud & Infrastructure: AWS (Lambda, EC2, S3, EMR, IAM), Azure (Blob, Data Factory, Security Center), GCP (BigQuery, Storage). CI/CD with GitHub Actions, Jenkins, Docker, Kubernetes, Terraform.

Databases: SQL & NoSQL (PostgreSQL, Cassandra, MongoDB, Databricks). Advanced in Python, Scala, SQL, and Java for analytics & automation.

Machine Learning & Analytics: TensorFlow, PyTorch, SageMaker; model training, tracking, and deployment at scale.

Observability & Security: Monitoring with Prometheus, Grafana, CloudWatch, ELK, Datadog; service mesh with Linkerd; strong grounding in PCI/GDPR compliance.

LANGUAGES

English — ***Native speaker***

French — ***Working knowledge***

CERTIFICATIONS

Certified Kubernetes Administrator

AWS Certified Data Analytics - Specialty

Tableau Desktop Certified Professional